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Case 3-2021: A 48-Year-Old Man with Transient Vision Loss

Sebastian H. Unizony, M.D., Hillary R. Kelly, M.D., Elizabeth K. O'Donnell, M.D.,
and Sam Sadigh, M.D.

PRESENTATION OF CASE

Dr. Naomi J. Serling-Boyd (Medicine): A 48-year-old man was evaluated at this hospital because of transient vision loss.

The patient had been in his usual state of good health until 1 hour before evaluation, when he noticed white “sparkles” flickering in the vision of the left eye, followed by sudden blurriness. He had no floaters or eye pain. During the next 2 minutes, the blurry vision worsened and became “dark,” such that the patient could not see a hand in front of his face when he covered the right eye. He presented to this hospital for evaluation, arriving 1 hour after the onset of vision symptoms.

On presentation, the patient noted that the vision in the left eye was improving but remained blurry. The temperature was 36.7°C, the blood pressure 120/65 mm Hg, the heart rate 92 beats per minute, and the body-mass index (the weight in kilograms divided by the square of the height in meters) 24.2. Visual acuity was 20/20 in the right eye and was limited to hand motions at 0.9 m (3.0 ft) in the left eye; 2 hours later, visual acuity in the left eye was 20/40 for distance vision and 20/30 with correction for near vision. Results of the Ishihara color test were normal. The pupils were equal, round, and reactive to light, with no relative afferent pupillary defect. Extraocular movements were intact; visual fields were full in response to confrontation. Intraocular pressure and results of a slit-lamp examination of the anterior and posterior segments of the eye were normal bilaterally. There was severe ptosis of the left upper eyelid (margin reflex distance, 4 mm in the right eye and 1 mm in the left eye). A dilated fundus examination revealed normal optic disks, maculae, and vessels. A nontender, irregular mass was palpable under the left superior orbital rim, but there was no proptosis. The temporal arteries were prominent bilaterally, more so on the right side than on the left side, with normal pulsations and no tenderness. There was no tenderness of the scalp or temporomandibular joints. The remainder of the physical examination was normal.

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Table 1. Laboratory Data.*

Variable	Reference Range†	On Admission
White-cell count (per μ l)	4500–11,000	6310
Differential count (per μ l)		
Neutrophils	1800–7700	4520
Lymphocytes	1000–4800	1160
Monocytes	200–1200	440
Eosinophils	0–900	170
Basophils	0–300	20
Hemoglobin (g/dl)	13.5–17.5	11.6
Hematocrit (%)	41.0–53.0	35.0
Platelet count (per μ l)	150,000–400,000	258,000
Erythrocyte sedimentation rate (mm/hr)	0–13	53
C-reactive protein (mg/liter)	<8.0	11.8
Alanine aminotransferase (U/liter)	10–55	245
Aspartate aminotransferase (U/liter)	10–40	154
Alkaline phosphatase (U/liter)	45–115	1924
Bilirubin (mg/dl)		
Total	0.0–1.0	2.8
Direct	0.0–1.0	0.0
Protein (g/dl)		
Total	6.0–8.3	7.6
Albumin	3.3–5.0	4.2
Globulin	1.9–4.1	3.4
Cholesterol (mg/dl)		
Total	<200	280
Low-density lipoprotein	50–129	168

* To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for cholesterol to millimoles per liter, multiply by 0.02586.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

A review of systems was notable for fatigue and weight loss of 4.5 kg in the past year, which the patient attributed to inactivity and decreased appetite. There was a 2-month history of bilateral jaw pain with chewing of firm foods but not with soft foods. There was lateral hip soreness in both hips after walking but no stiffness. There was a sensation of locking when flexing or extending the right first finger. Two days before evaluation, the patient had a temporal headache on the right side that resolved. He reported no numbness or weakness.

The patient had a history of presbyopia. He

had not undergone any surgeries, was not currently taking any medications, and had no known allergies. He was born and raised in a country in Eurasia and had lived in the United States for 20 years. He traveled internationally for his work as a university professor. He was divorced and lived with his two children in a house in New England. He had smoked a half pack of cigarettes for 5 years but had quit smoking 20 years earlier. He drank alcohol rarely and did not use illicit drugs. His parents were alive and well, both of his grandmothers had had breast cancer, and there was no family history of ocular or autoimmune disease.

Laboratory evaluation revealed normocytic anemia and elevated levels of alanine aminotransferase, aspartate aminotransferase, and alkaline phosphatase. Results of tests of kidney and thyroid function were normal, as were blood levels of electrolytes, calcium, glucose, glycated hemoglobin, creatine kinase, homocysteine, and lipoprotein(a). Testing for antinuclear antibodies was negative, and levels of IgG subclasses were normal. Other laboratory test results are shown in Table 1. Aspirin was administered, along with empirical treatment with intravenous methylprednisolone, and the patient was admitted to the hospital.

Dr. Hillary R. Kelly: Magnetic resonance imaging (MRI) of the head and orbits (Fig. 1), performed before and after the administration of intravenous contrast material, revealed enlargement of the right medial rectus muscle with stranding of the adjacent extraconal fat, as well as an enhancing masslike lesion in the left anterior superior orbit, abutting the left lacrimal gland. The masslike lesion showed a hypointense signal on short-tau inversion recovery imaging. The brain parenchyma was normal. Magnetic resonance angiography of the head and neck was normal, as was ultrasonography of the temporal arteries. Ultrasonography of the abdomen revealed markedly heterogeneous liver parenchyma and an enlarged periportal lymph node, measuring 2 cm in largest dimension. Computed tomography (CT) of the chest, abdomen, and pelvis (Fig. 1), performed after the administration of intravenous contrast material, revealed multiple punctate calcifications throughout the liver and hypoattenuation of the right hepatic lobe, as well as diffuse coalescent mesenteric and retroperitoneal lymphadenopathy with coarse

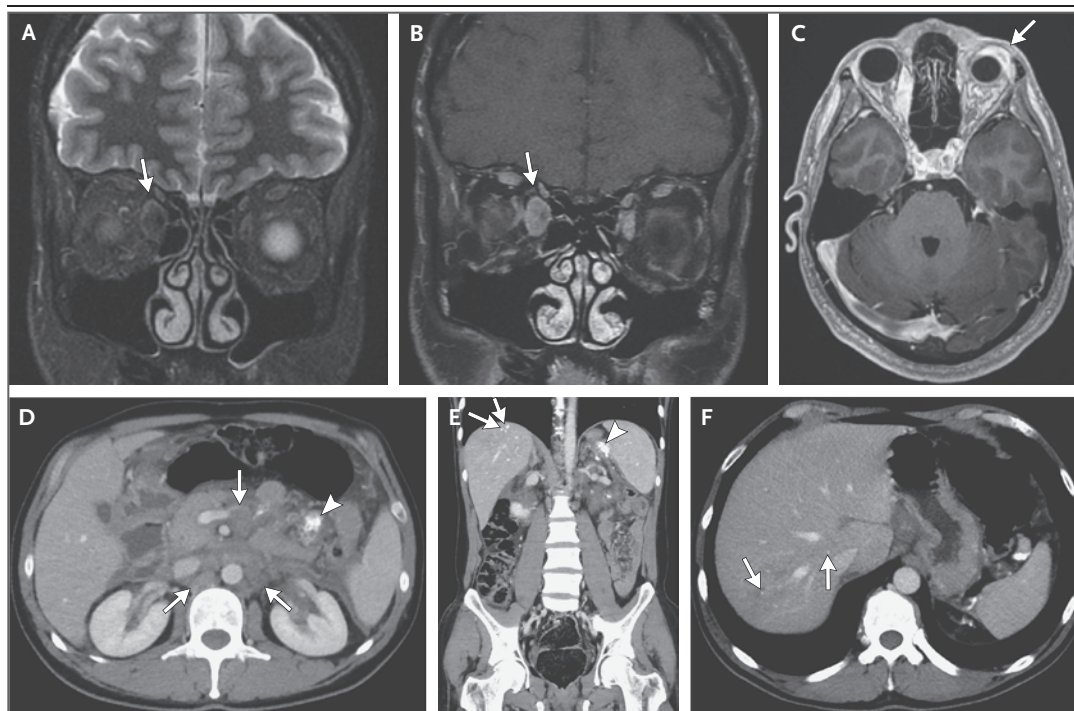


Figure 1. Imaging Studies.

MRI of the head and orbits was performed. A coronal short-tau inversion recovery image (Panel A) and a coronal T1-weighted fat-saturated image obtained after the administration of intravenous gadolinium (Panel B) show enlargement of the right medial rectus muscle (arrows), which is associated with marked T2-hypointense signal (Panel A) but shows enhancement similar to that of the other extraocular muscles (Panel B). Mild enhancement and fat stranding are present in the adjacent extraconal fat. An axial three-dimensional fast spoiled gradient echo image (Panel C), obtained after the administration of intravenous gadolinium, shows an enhancing masslike lesion (arrow) in the left anterior superior orbit, abutting the left lacrimal gland. CT of abdomen and pelvis was also performed. Axial and coronal images (Panels D and E, respectively), obtained after the administration of intravenous contrast material, show diffuse coalescent mesenteric and retroperitoneal lymphadenopathy (Panel D, arrows). An additional axial image (Panel F) shows geographic areas of heterogeneous hypoattenuation in the right lobe of the liver. Multiple small calcifications of varying shapes and sizes are shown throughout the liver (Panels E and F, arrows). In addition, coarse calcifications are shown within some of the mesenteric and retroperitoneal lymph nodes (Panels D and E, arrowheads).

calcifications. There was a prominent right hilar lymph node, measuring 10 mm in diameter. The lung parenchyma was unremarkable.

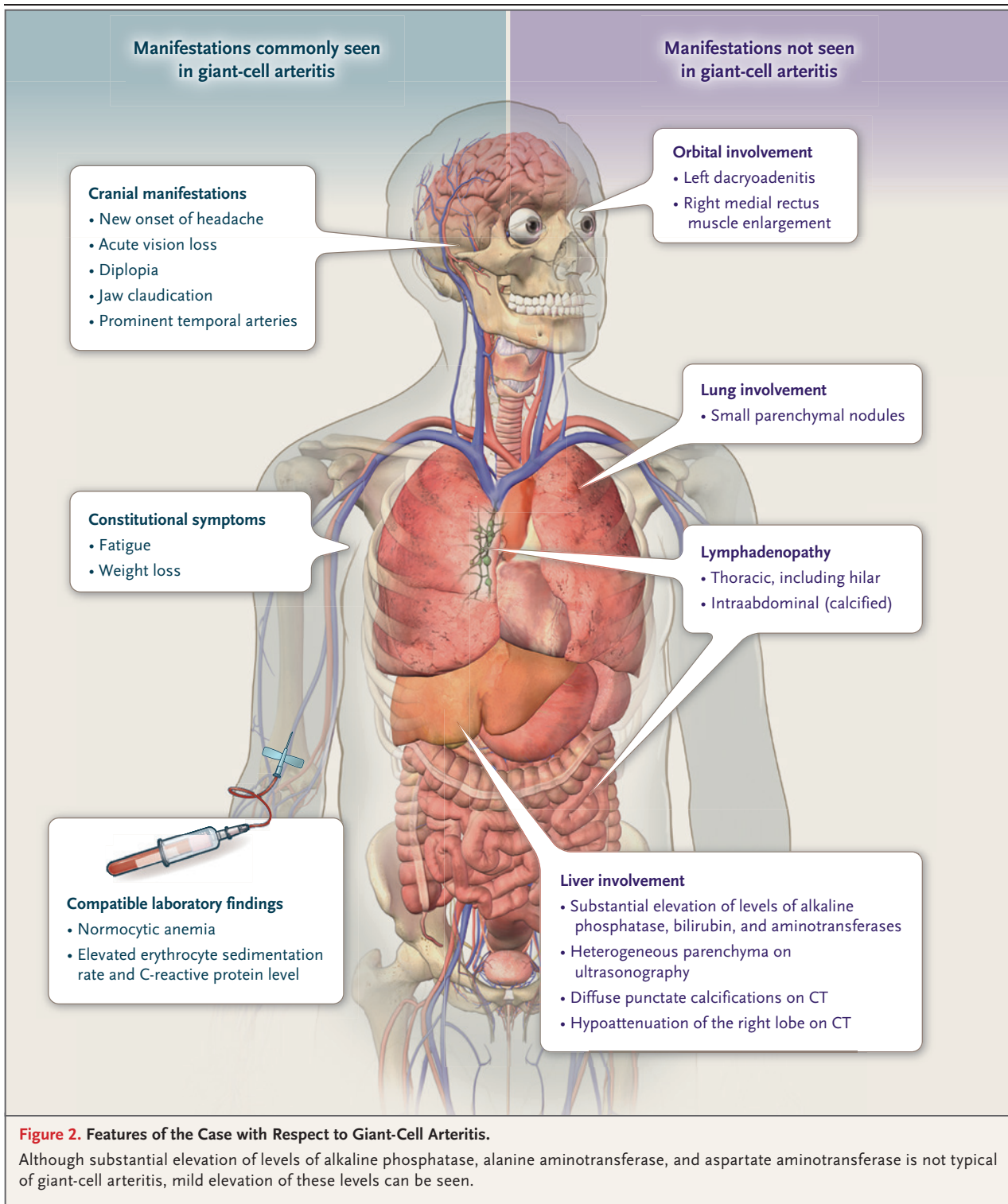
Dr. Serling-Boyd: On the second hospital day, diplopia occurred when the patient looked to the far right. On examination, there was mild limitation of abduction of the right eye. After a second dose of intravenous methylprednisolone was administered, the patient chose to leave the hospital against medical advice. After discharge, he took prednisone daily. During the next week, additional laboratory test results became available. Testing for antineutrophil cytoplasmic antibodies (ANCA) was negative. Blood testing for blastomyces antigen, coccidioides antibodies, and histoplasma antibodies was negative, as was an

interferon- γ release assay for *Mycobacterium tuberculosis*.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Sebastian H. Unizony: This patient has several clinical manifestations that raise concern about giant-cell arteritis (Fig. 2), even if he is younger than the typical patient presenting with this disease. These manifestations include acute vision loss, new onset of headache, mandibular pain with mastication (also referred to as jaw claudication), diplopia, and constitutional symptoms.¹ In addition, the physical examination is notable for prominent temporal arteries, a finding that



is sometimes seen in patients with giant-cell arteritis, and the laboratory evaluation is notable for an elevated erythrocyte sedimentation rate and

C-reactive protein level, abnormalities that are present in nearly 95% of patients with giant-cell arteritis before glucocorticoids are administered.²

In this case, after the treating physicians encountered a clinical syndrome compatible with giant-cell arteritis with the knowledge that irreversible blindness occurs in up to 20% of patients with this disease (nearly always before the diagnosis is confirmed),¹ they appropriately initiated glucocorticoid therapy without delay, before the workup was complete. However, several features of this patient's presentation are not consistent with giant-cell arteritis (Fig. 2), triggering suspicion for conditions that mimic this diagnosis.³ He has orbital abnormalities, including mass effect due to left dacryoadenitis, as well as enlargement of the right medial rectus muscle, which probably explains his diplopia. In addition, he has enlarged, calcified lymph nodes in the abdomen and a prominent lymph node in the chest. Blood tests show substantial elevation of levels of alkaline phosphatase, bilirubin, and aminotransferases. Finally, imaging studies of the liver show heterogeneous parenchyma with diffuse punctate calcifications and hypoattenuation of the right lobe, findings that are suggestive of an infiltrative process.

DISEASES MIMICKING GIANT-CELL ARTERITIS

My differential diagnosis for this constellation of clinical manifestations — including cranial symptoms resembling giant-cell arteritis, dacryoadenitis, extraocular muscle enlargement, lymphadenopathies, and liver infiltration — includes sarcoidosis, IgG4-related disease, and primary amyloidosis. Vasculitides involving small and medium vessels (e.g., polyarteritis nodosa and ANCA-associated vasculitis) sometimes affect the head and neck arteries and can be confounded with giant-cell arteritis.³ However, these vasculitides do not usually involve the liver parenchyma or the lymph nodes. In addition, patients with these vasculitides commonly have other clinical manifestations (e.g., cutaneous, otorhinolaryngologic, pulmonary, peripheral nerve, or renal manifestations) that facilitate disease recognition, and these findings are absent in this case. Furthermore, 40 to 90% of patients with ANCA-associated vasculitis have a positive ANCA test, and testing for ANCAs was negative in this case. Tuberculosis and endemic mycosis (e.g., histoplasmosis, coccidioidomycosis, and blastomycosis) can involve the liver and lymph nodes and can be associated with some of the imaging features observed in this patient. However, these

infections do not cause symptoms consistent with giant-cell arteritis, and corresponding microbiologic tests were negative in this case.

SARCOIDOSIS

Sarcoidosis is an immune-mediated disorder of unknown cause that can affect all racial groups. It can occur anytime in adulthood but usually develops before 50 years of age. The disease is characterized by noncaseating granulomatous inflammation that can involve any organ.⁴ Approximately 90% of patients present with intrathoracic abnormalities such as hilar lymphadenopathy and interstitial lung disease.⁵ In approximately 30% of cases, extrathoracic manifestations occur, most commonly involving the joints (e.g., arthralgias), the eyes (e.g., uveitis), and the skin (e.g., erythema nodosum).⁵ Constitutional symptoms are frequently reported. Sarcoidosis can also affect the central and peripheral nervous system, extraocular orbital structures (e.g., lacrimal glands and extraocular muscles), liver, spleen, heart, gastrointestinal tract, kidneys, and calcium metabolism.⁵ In rare cases, sarcoidosis can be complicated by systemic vasculitis affecting small, medium, or large vessels.⁶ In fact, sarcoidosis-associated vasculitis involving the extracranial arteries (e.g., temporal arteries) with resultant headache and jaw claudication has been reported.⁷⁻⁹

This patient could have sarcoidosis with an atypical presentation that mimics giant-cell arteritis, in addition to infiltration of the lacrimal glands, extraocular muscles, lymph nodes, and liver. Autopsy studies have shown that liver involvement occurs in most patients with sarcoidosis but becomes clinically significant in approximately 10 to 30% of patients.¹⁰ Hepatitis associated with sarcoidosis sometimes causes parenchymal hypoattenuation and leads to dystrophic calcification, findings that are seen in this patient. Arguing against the possibility of sarcoidosis, however, is the fact that this patient has a single prominent right hilar lymph node measuring 10 mm in diameter; in patients with sarcoidosis, hilar lymphadenopathy tends to be associated with larger, multiple, and bilateral lymph nodes.⁴ Measurement of angiotensin-converting enzyme and 1,25-dihydroxyvitamin D levels could be helpful in this case; elevated levels would support but not confirm the diagnosis of sarcoidosis. The most appropriate con-

firmatory test would be a tissue biopsy (e.g., lymph node, lacrimal gland, or liver biopsy), with the specimen showing noncaseating granulomas.

IGG4-RELATED DISEASE

IgG4-related disease is a multisystem fibroinflammatory disease that has been increasingly recognized over the past decade.¹¹ Men are affected more frequently than women, and the age at onset is typically between 50 and 60 years. Lesions associated with IgG4-related disease are characterized by lymphoplasmacytic infiltrates and storiform fibrosis.¹¹ The most commonly affected organs are the pancreas (i.e., autoimmune pancreatitis type 1), the hepatobiliary system (e.g., sclerosing cholangitis), and the salivary and lacrimal glands.¹² Other possible manifestations include lymphadenitis, dermatitis, thyroiditis, pachymeningitis, hypophysitis, pericarditis, tubulointerstitial nephritis, lung parenchymal masses, orbital pseudotumor formation, and extraocular muscle involvement.¹² Vasculitis such as aortitis with periaortitis, sometimes concurrent with retroperitoneal fibrosis, is another potential manifestation of IgG4-related disease,¹³ and temporal arteritis has been reported occasionally.^{14,15}

Patients with IgG4-related disease present with synchronous involvement of three organs, on average. The main clinical phenotypes of organ involvement are hepatopancreatobiliary disease, retroperitoneal fibrosis or aortitis, disease limited to the head and neck, and classic Mikulicz syndrome with systemic features.¹⁶ Patients with disease limited to the head and neck or with classic Mikulicz syndrome with systemic features tend to have the highest serum IgG4 levels and frequently have inflammation of the lacrimal glands (in 50 to 60% of cases), salivary glands (50 to 80%), and lymph nodes (30 to 70%); however, these two clinical phenotypes are not frequently associated with involvement of the liver (in <5% of cases) or biliary tree (1 to 30%).¹⁶

Although this patient could have an uncommon form of IgG4-related disease, several features of his presentation are not consistent with this diagnosis. First, patients with IgG4-related disease tend to have inflammation of both the lacrimal glands and the salivary glands bilaterally,^{11,16,17} whereas this patient has unilateral lacrimal gland enlargement and no obvious salivary gland disease. Second, many patients with

IgG4-related disease have pancreatic involvement,^{11,16} which is absent in this patient. Third, most patients with IgG4-related disease have normal serum inflammatory marker levels and increased serum IgG4 levels (>135 mg per deciliter),¹⁸ especially in the presence of multiorgan disease and infiltration of exocrine glands, and these laboratory findings are absent in this patient. Fourth, this patient has lymph node and liver calcifications, which are not commonly observed in patients with IgG4-related disease. Finally, reports of IgG4-related disease with cranial manifestations that mimic giant-cell arteritis are rare.

SYSTEMIC AMYLOIDOSIS

Systemic amyloidosis is a rare disease that results from protein misfolding and aggregation into highly ordered fibrils that deposit in tissues and cause progressive organ damage. Amyloidosis can occur in the presence of an abnormal protein (e.g., primary amyloidosis, also known as amyloid light-chain [AL] amyloidosis), in association with excess abundance of a normal protein (e.g., secondary amyloidosis, also known as amyloid A [AA] amyloidosis), and for unclear reasons (e.g., senile systemic amyloidosis).¹⁹

Primary amyloidosis is a plasma-cell dyscrasia caused by deposition of protein fragments derived from abnormal immunoglobulin light chains.¹⁹ The mean age at diagnosis is 63 years, and men are affected more often than women.¹⁹ In most patients with primary amyloidosis, the amyloidogenic light chains are produced by low-level expansion of an indolent B-cell clone that arises in the context of monoclonal gammopathy of undetermined significance. In some patients, the disease occurs in association with multiple myeloma, Waldenström's macroglobulinemia, or non-Hodgkin's lymphoma. The most frequently affected organs are the kidneys (leading to nephrotic proteinuria) and the heart (causing pump failure). Patients may also have peripheral and autonomic neuropathy, joint and soft-tissue infiltration (e.g., extraocular muscle enlargement),²⁰ deposits in the lacrimal glands²¹ and salivary glands, gastrointestinal dysmotility, skin lesions (e.g., papules and ecchymosis), lymphadenopathy,²² splenomegaly, liver disease, respiratory manifestations (e.g., lung nodules), and bleeding diathesis.¹⁹ Liver involvement, which occurs in approximately 20 to 30% of cases of

primary amyloidosis,²³ may manifest as elevated aminotransferase levels, cholestasis, and hepatomegaly. CT of the liver may show areas of parenchymal hypoattenuation in patients with clinically significant amyloid deposition.²⁴ Furthermore, amyloid fibrils have an affinity for calcium, and imaging studies can show calcified deposits in different organs, including the liver²⁴ and lymph nodes,²⁵ findings similar to those seen in this patient.

Primary amyloidosis can affect any organ, including the arteries, except for the brain. In fact, several cases of primary amyloidosis with cranial features similar to those of giant-cell arteritis have been reported.^{20,26-31} The diagnosis of primary amyloidosis may well explain this patient's constellation of clinical manifestations, including the liver involvement with characteristic findings on imaging (parenchymal hypoattenuation and calcification), lymphadenopathy, lacrimal gland involvement, extraocular muscle enlargement, and a syndrome resembling giant-cell arteritis, with prominent temporal arteries, temporal headache, jaw claudication, and transient vision loss. In cases of amyloidosis mimicking giant-cell arteritis, vision loss is presumably caused by transient or permanent occlusion of the posterior ciliary arteries or the retinal arteries.^{20,29,32} In addition, this patient reports lateral hip soreness and symptoms suggestive of stenosing flexor tenosynovitis (trigger finger) of the right first finger, which could be caused by amyloid infiltration in soft tissues. Finally, the increase in the erythrocyte sedimentation rate is disproportionate to the increase in the C-reactive protein level, a finding that perhaps reflects the presence of paraproteinemia, which is seen in more than 90% of patients with primary amyloidosis.

The diagnosis of amyloidosis requires the identification of amyloid fibrils in tissue. On microscopic examination, these fibrils show a pathognomonic apple-green birefringence under polarized light after Congo red staining.¹⁹ To establish the diagnosis in this case, I would recommend performing immunoelectrophoresis and quantifying free light chains in blood and urine to look for a monoclonal component, and most important, performing a biopsy to confirm the presence of amyloid deposits. A fat-pad biopsy is simple to perform and has a relatively high yield for the diagnosis of systemic primary

amyloidosis. Other tissues that could be sampled in this patient include the rectal or lip mucosa, left lacrimal gland, temporal arteries, liver, and abdominal lymph nodes. Because amyloidosis is often associated with hemostatic defects, less invasive procedures are generally recommended as an initial diagnostic approach. If amyloid material is identified, the next steps are to determine the type of amyloid with the use of immunohistochemical studies or laser-capture microdissection followed by mass spectrometry, as well as to assess the extent and severity of the disease.¹⁹ Where available, imaging with radioiodinated serum amyloid P component can be used to measure the burden of disease and monitor the response to treatment.³³ If the diagnosis of primary amyloidosis is confirmed in this patient, I would also recommend a bone marrow biopsy to rule out multiple myeloma, which occurs in 10 to 15% of patients with primary amyloidosis.

DR. SEBASTIAN H. UNIZONY'S
DIAGNOSIS

Systemic primary amyloidosis.

PATHOLOGICAL DISCUSSION

Dr. Sam Sadigh: The diagnostic tests in this case were biopsies of a retroperitoneal lymph node and the liver. Histologic examination of the retroperitoneal lymph node revealed fragments of fibroadipose tissue with calcifications and acellular eosinophilic deposits in areas with cracking artifact (Fig. 3A). Associated scattered multinucleated giant cells and reactive osseous metaplasia were present. No identifiable lymph node tissue was seen. On Congo red staining, the eosinophilic deposits showed apple-green birefringence under polarized light, a feature characteristic of amyloid (Fig. 3B).

Histologic examination of the liver specimen revealed predominantly portal-based, well-delimited nodules of similar eosinophilic deposits (Fig. 3C), which were consistent with amyloid (Fig. 3D). Foci adjacent to the amyloid deposits in the liver contained a cellular infiltrate with small but dense clusters of mature-appearing plasma cells (Fig. 3E). On immunohistochemical staining for kappa and lambda light chains, the plasma cells showed kappa restriction (Fig. 3F

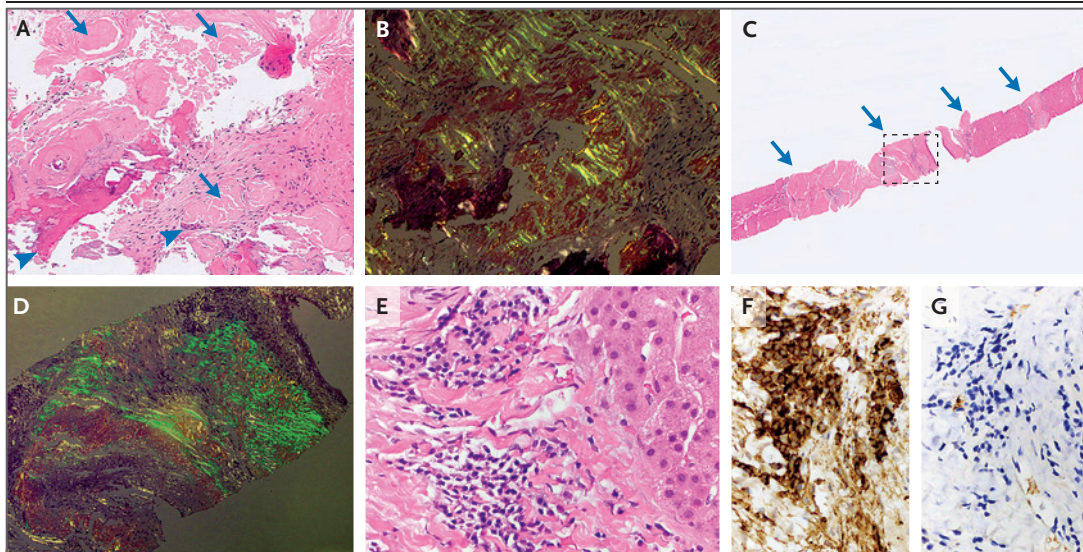


Figure 3. Biopsy Specimens of a Lymph Node and the Liver.

Hematoxylin and eosin staining of a biopsy specimen of a retroperitoneal lymph node (Panel A) shows fragments of fibroadipose tissue with calcifications and acellular eosinophilic deposits (arrows), with associated reactive osseous metaplasia (left arrowhead) and scattered multinucleated giant cells (right arrowhead). On Congo red staining (Panel B), the eosinophilic deposits show apple-green birefringence under polarized light, a feature characteristic of amyloid. Hematoxylin and eosin staining of a biopsy specimen of the liver (Panel C) shows portal-based, well-delineated nodules of eosinophilic deposits (arrows); the deposits within the dashed outline show apple-green birefringence under polarized light on Congo red staining (Panel D). At higher magnification (Panel E), there is a focal dense cluster of plasma cells adjacent to the amyloid deposits in the liver parenchyma. On immunohistochemical staining for kappa and lambda light chains (Panels F and G, respectively), the plasma cells show kappa restriction.

and 3G), which indicated the presence of a plasma-cell neoplasm associated with the parenchymal amyloid deposition.

Even in patients with a plasma-cell neoplasm, it is still essential to characterize the type of amyloid, since secondary or familial amyloidosis may be present incidentally. Laser-capture microdissection of the amyloid in a tissue biopsy specimen and analysis by means of mass spectrometry is the most sensitive and specific method of subtyping amyloid. In this case, a section of the biopsy specimen of the retroperitoneal lymph node was analyzed by means of liquid chromatography with tandem mass spectrometry, which showed a peptide pattern consistent with kappa immunoglobulin light-chain amyloid.

A bone marrow biopsy was performed as part of the workup for potential plasma-cell myeloma. Examination of a core biopsy specimen revealed mildly hypocellular marrow with maturing tri-lineage hematopoiesis (Fig. 4A and 4B). However, scattered and clustered plasma cells were present that showed kappa restriction on kappa

and lambda light-chain mRNA in situ hybridization (Fig. 4C and 4D). On immunohistochemical staining for CD138, the plasma cells accounted for approximately 10% of the marrow cellularity (approximately 5% of the intertrabecular marrow space). Congo red staining was negative for amyloid deposits in the marrow.

In bone marrow aspirate, plasma cells accounted for 8.5% of the marrow cellularity on a differential count (Fig. 4E). On flow cytometry of the aspirate, plasma cells accounted for 3.7% of all viable events and revealed the following immunophenotype: CD45dim CD138+ CD38+ CD19dim/– CD27+ CD56– with expression of monotypic cytoplasmic kappa immunoglobulin light chain (Fig. 4F). Cytogenetic studies and fluorescence in situ hybridization of the bone marrow revealed a normal male karyotype (46,XY[20]); however, the translocation t(11;14) was identified, reflecting fusion of the *CCND1* and *IGH* loci. Although early studies suggested that the presence of this translocation at diagnosis might be associated with a favorable prognosis in patients with multiple myeloma, its pres-

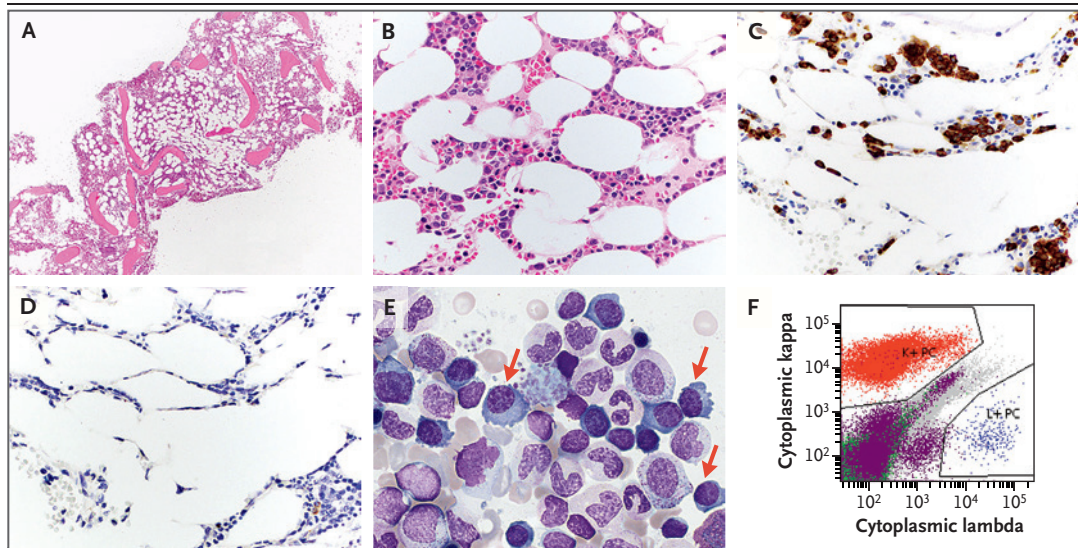


Figure 4. Biopsy Specimen of the Bone Marrow.

Hematoxylin and eosin staining of a core biopsy specimen of the bone marrow (Panel A) shows mildly hypocellular marrow for the patient's age with maturing trilineage hematopoiesis. At higher magnification (Panel B), scattered and clustered plasma cells are present. On kappa and lambda light-chain mRNA in situ hybridization (Panels C and D, respectively), the plasma cells show kappa restriction. On Wright–Giemsa staining of a smear of bone marrow aspirate (Panel E), the mature plasma cells (arrows) account for 8.5% of the marrow cellularity on a differential count. On flow cytometry of the aspirate (Panel F), the vertical and horizontal axes show fluorescence intensity measurements for cytoplasmic kappa and lambda light chains, respectively. There is a predominance of clonal kappa-positive plasma cells (K+ PC), as indicated by red dots, with very few lambda-positive plasma cells (L+ PC), as indicated by blue dots; purple and green dots indicate non-plasma cell populations.

ence may be prognostically adverse in patients with primary amyloidosis.^{34–36} The overall finding of monotypic plasma cells in the bone marrow in the presence of light-chain amyloid in lymph node and liver specimens on mass spectrometry confirmed the diagnosis of systemic primary amyloidosis.

ANATOMICAL DIAGNOSIS

Systemic primary amyloidosis.

DISCUSSION OF MANAGEMENT

Dr. Elizabeth K. O'Donnell: For the treatment of primary amyloidosis with hepatic involvement, the patient received daratumumab, bortezomib, and dexamethasone. Daratumumab is a human IgG1k monoclonal antibody that targets CD38, a cell-surface glycoprotein highly expressed on plasma cells; bortezomib is a proteasome inhibitor; and dexamethasone is a glucocorticoid. From initiation to completion of three cycles of therapy, the alkaline phosphatase level decreased from 1764 U

per liter to 939 U per liter, the alanine aminotransferase level from 457 U per liter to 190 U per liter, the aspartate aminotransferase level from 435 U per liter to 121 U per liter, and the serum level of kappa free light chains from 1271.4 mg per liter to 78.3 mg per liter.

The patient underwent a total of four cycles of therapy and has not had any clinically significant side effects. He is now working full time and reports that he feels well.

FINAL DIAGNOSIS

Systemic primary amyloidosis.

This case was presented at the Medicine Case Conference.

Dr. Unizony reports receiving consulting fees from Kiniksa and Janssen and research support from Genentech; Dr. Kelly, receiving clinical trial funding, paid to her institution, from Bayer; and Dr. O'Donnell, receiving clinical trial funding, paid to her institution, consulting fees, and honoraria from Janssen and Bristol Myers Squibb, clinical trial funding, paid to her institution, from Amgen, clinical trial funding, paid to her institution, and honoraria from Takeda, and honoraria from Karyopharm and Oncopeptides. No other potential conflict of interest relevant to this article was reported.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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